

AWS RDS Connectivity

Summary: We use Amazon Web Services RDS (Relational Database Service) connectivity to allow applications, websites, or servers such as Amazon EC2 instances to securely connect with a managed database like MySQL, PostgreSQL, or Oracle. It helps applications store, retrieve, and manage data without manually handling database server maintenance, backups, updates, or scaling.

The need to modify RDS connectivity settings instead of using default configurations is mainly for:

- **Security** – Default settings may not be secure for real projects. We modify security groups, ports, and public/private access to control who can access the database.
- **Performance** – Applications may require optimized storage, memory, or connection settings for faster response.
- **Scalability** – Default configurations may not handle high traffic, so settings are changed for better scaling and load handling.
- **Availability** – Features like Multi-AZ deployment and automated backups are configured to improve reliability.
- **Customization** – Different projects need different database engines, parameter groups, and networking setups.
- **Cost Optimization** – Default resources may be unnecessary or expensive, so configurations are adjusted based on project needs.

Create database [Info](#)

Availability zone group [Info](#)

Availability zone group

Select the availability zone where you want to create your database.

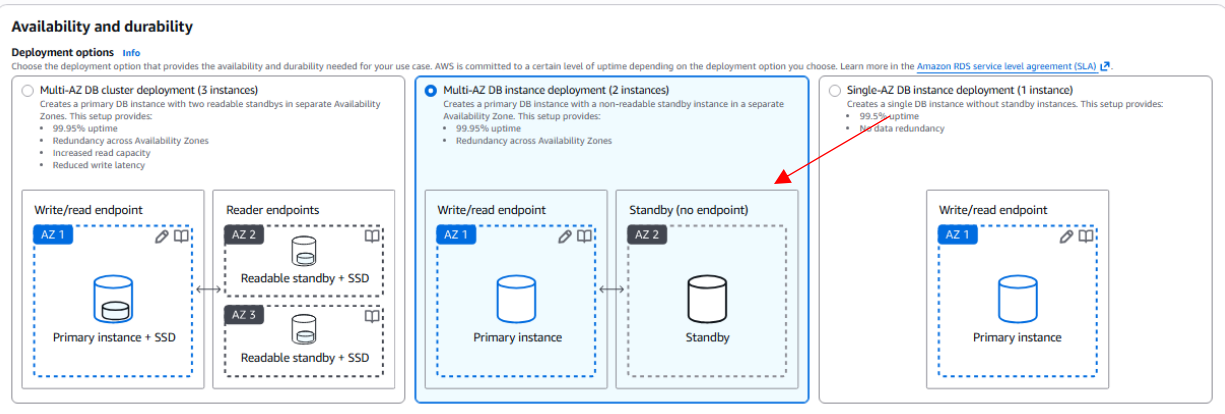
US East (N. Virginia) ▼

Engine options

Engine type [Info](#)

☐ Aurora (MySQL Compatible) 	☐ Aurora (PostgreSQL Compatible) 	☒ MySQL 	☐ PostgreSQL 
☐ MariaDB 	☐ Oracle 	☐ Microsoft SQL Server 	☐ IBM Db2 

- To set up connectivity, we first create a database instance in RDS.



- For this guide, we are using the **Multi-AZ DB instance deployment (2 instances)** deployment option. This configuration automatically provisions and manages two database instances across two separate Availability Zones, providing high availability and automatic failover.

Connectivity Info

Compute resource

Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.

☒ **Don't connect to an EC2 compute resource**

Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.

☐ **Connect to an EC2 compute resource**

Set up a connection to an EC2 compute resource for this database.

Virtual private cloud (VPC) Info

Choose the VPC. The VPC defines the virtual networking environment for this DB instance.

Default VPC (vpc-099792461c7d0a616)
6 Subnets, 6 Availability Zones

After a database is created, you can't change its VPC. Only VPCs with a corresponding DB subnet group are listed.

DB subnet group Info

Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.

default-vpc-099792461c7d0a616
6 Subnets, 6 Availability Zones

Public access Info

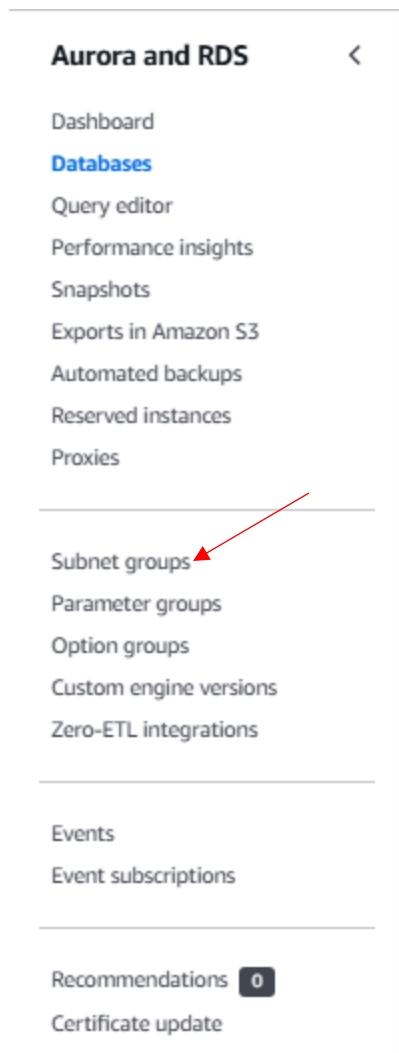
☐ **Yes**

RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.

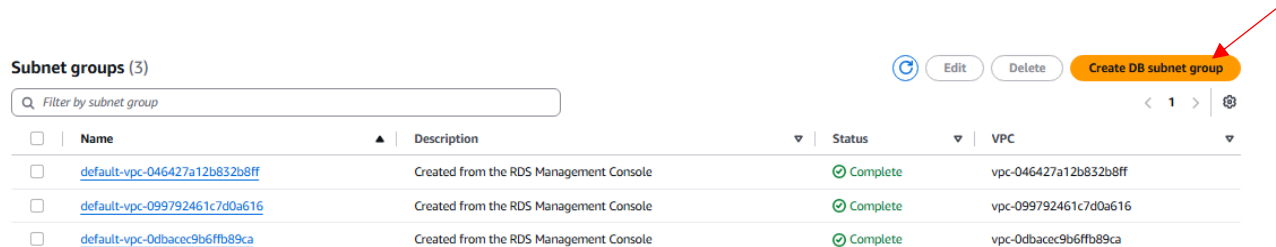
☒ **No**

RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database.

- Select the VPC (Virtual Private Cloud) and DB subnet group. The VPC defines the isolated network environment in which your database will reside, while the DB subnet group specifies which subnets within that VPC the database can use.
- Before creating the database, you must first create a DB subnet group. A DB subnet group is a collection of subnets (in different Availability Zones) that AWS uses to place your RDS instances.



- In the RDS dashboard, navigate to the “Subnet groups” section and click “Create DB subnet group” to begin creating your subnet group.



- AWS provides some default subnets, but for this setup we will create a new custom subnet group to have better control over networking.

Subnet group details

Name
You won't be able to modify the name after your subnet group has been created.
MyDBSubnetGroup

Description
Must contain from 1 to 255 characters. Alphanumeric characters, spaces, hyphens, underscores, and periods are allowed.

VPC
Choose a VPC identifier that corresponds to the subnets you want to use for your DB subnet group. You won't be able to choose a different VPC identifier after your subnet group has been created.
Default VPC (vpc-099792461c7d0a616)
6 Subnets, 6 Availability Zones

- Enter a subnet name, add a description, and select the appropriate VPC from the dropdown list.

Add subnets

Availability Zone group info
Choose the Availability Zone group that includes the subnets that you want to add. An Availability zone group includes a set of Availability Zones.
us-east-1-zg-1

Availability Zones
Choose the Availability Zones that include the subnets you want to add.
Choose an availability zone
us-east-1a us-east-1b

Subnets
Choose the subnets that you want to add. The list includes the subnets in the selected Availability Zones.
Select subnets
Subnet ID: subnet-0b5783ef03796c671 CIDR: 172.31.16.0/20 Subnet ID: subnet-0d10d67b324b6c997 CIDR: 172.31.32.0/20

① For Multi-AZ DB clusters, you must select 3 subnets in 3 different Availability Zones.

Subnets selected (2)

Availability zone	Subnet name	Subnet ID	CIDR block
us-east-1a	-	subnet-0b5783ef03796c671	172.31.16.0/20
us-east-1b	-	subnet-0d10d67b324b6c997	172.31.32.0/20

Cancel Create

- Next, select the Availability Zones. Availability Zones are physically separate data centers within an AWS region. Placing your subnets in different zones ensures redundancy.
- Note: You must select at least 2 Availability Zones because we are creating a database using the **Multi-AZ DB instance deployment (2 instances)**. Each instance is placed in a different zone, which is required for failover support.
- Select a specific subnet ID for each chosen Availability Zone. Each subnet ID corresponds to a defined IP address range within that zone.
- Click “Create” to finalize the DB subnet group. It will now be available for use when configuring your RDS database.

DB subnet group Info
Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.

mysubnet
2 Subnets, 2 Availability Zones

default-vpc-099792461c7d0a616
6 Subnets, 6 Availability Zones

mysubnet
2 Subnets, 2 Availability Zones

- Return to the Database creation page in the RDS console to continue with the remaining configuration steps.

- The subnet group you just created will now appear in the “DB subnet group” dropdown on the database creation page.
- Select the newly created subnet group from the list to associate it with your database.

Existing VPC security groups

Choose one or more options

Q

- ☒ EC2-SG
- ☐ ElasticMapReduce-master
- ☐ default
- ☐ security-group-for-inbound-nfs-d-fp6dgesdqa8q
- ☐ RDS-SG
- ☐ security-group-for-outbound-nfs-d-zwr77jqzsylic
- ☐ launch-wizard-1
- ☐ security-group-for-outbound-nfs-d-fp6dgesdqa8q
- ☐ security-group-for-inbound-nfs-d-zwr77jqzsylic
- ☐ ElasticMapReduce-slave

- Set the **Existing VPC security groups** by selecting an existing VPC security group from the list. A security group acts as a virtual firewall, controlling which inbound and outbound traffic is allowed to reach your database. If you do not have one already, you can create a new security group through the VPC console before proceeding.

Certificate authority - optional [Info](#)

Using a server certificate provides an extra layer of security by validating that the connection is being made to an Amazon database. It does so by checking the server certificate that is automatically installed on all databases that you provision.

rds-ca-rsa4096-g1
Expiry: May 26, 2121

rds-ca-ecc384-g1
Expiry: May 26, 2121

rds-ca-rsa4096-g1
Expiry: May 26, 2121

rds-ca-rsa2048-g1 (default)
Expiry: May 26, 2061

- Also select a **Certificate authority** to enable SSL/TLS encryption for data in transit. This ensures that all communication between your application and the RDS instance is encrypted, protecting sensitive data from interception.